Hydro-Turbine Optimization

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7 August 2024

Abstract

This paper presents an optimization model for maximizing power output in a hydroelectric power station equipped with three distinct turbines. Each turbine's power generation is characterized by a quadratic function dependent on water flow rate, derived from experimental data and Bernoulli's equation. The primary objective is to determine the optimal allocation of incoming water flow among the turbines to achieve maximum total power generation for any given overall flow rate. A Lagrange multiplier-based optimization approach is employed to solve this constrained optimization problem, considering the turbines' individual flow rate limitations. The results demonstrate the effectiveness of the proposed method in significantly enhancing power generation compared to equal water distribution strategies. The findings offer valuable insights for optimizing the operation of hydroelectric power stations with multiple turbines.

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Application of Lagrange Multipliers in Optimizing Hydro-Electric Turbines

For the sake of simplicity, the problem has given us 3 turbines to deal with which generally follow a simple pattern. Turbines 1, 2 and 3 are given to us which produce KW1, KW2 and KW3 amounts of power which also allow Qt, Qt and Qt amounts of water to flow through them, given as cubic feet and our t being time in seconds. In essence the units we will be dealing with are cubic feet per second. However, the flows of the turbine are constrained by the values of Qt, Qt and Qt, so we must account for these limitations.

Procedure 1

We need to calculate Qnt for turbines 1, 2 and 3 to see the maximum flow each turbine can handle. The KW values are given as follows:

$$
\begin{aligned}
KW_1 &= (-18.89 + 0.1277Q_1 - 4.08 \cdot 10^{-5}Q_1^2)(170 - 1.6 \cdot 10^{-6}Q_T^2) \\
KW_2 &= (-24.51 + 0.1358Q_2 - 4.69 \cdot 10^{-5}Q_2^2)(170 - 1.6 \cdot 10^{-6}Q_T^2) \\
KW_3 &= (-27.02 + 0.1380Q_3 - 3.84 \cdot 10^{-5}Q_3^2)(170 - 1.6 \cdot 10^{-6}Q_T^2)
\end{aligned}
$$

Firstly, to find the maximum total energy productions, we will use KW1 + KW2 + KW3, keeping in mind the constraints of each turbine. We will firstly find the constraints of Q_1 , Q_2 and Q_3 .

$$0.1277 - (8.16 \cdot 10^{-5})Q_1(170 - 1.6 \cdot 10^{-6}Q_T^2)$$

$$Q_2 = [0.1358 - (9.38 \cdot 10^{-5})Q_2](170 - 1.6 \cdot 10^{-6}Q_T^2)$$

$$Q_3 = [0.1380 - (7.68 \cdot 10^{-5})Q_3](170 - 1.6 \cdot 10^{-6}Q_T^2)$$

Since our partials are equal to the value of λ we can solve Q_2 and Q_3 in terms of Q_1 .

$$Q_2 = 86.354 + 0.8699Q_1$$
$$Q_3 = 134.115 + 1.0625Q_1$$

Finally, we can use backwards substitution from the system of equations to calculate the exact value of Q_1 , instead using QT as our parameter for all 3 equations

$$Q_1 = 0.341Q_T - 75.134$$
$$Q_2 = 0.297Q_T + 20.995$$
$$Q_3 = 0.362Q_T + 54.285$$

Procedure 2

Now, to calculate the validity of QT, we will look at the values for each of the turbines. We need to maximize Q_1 first.

$$Q_1 = .341Q_T - 75.134 \leq 1110 \quad Q_1 \leq 3475$$

Followed by maximizing Q_2 .

$$Q_2 = .297Q_T + 20.995 \leq 1110 \quad Q_2 \leq 3671.628$$

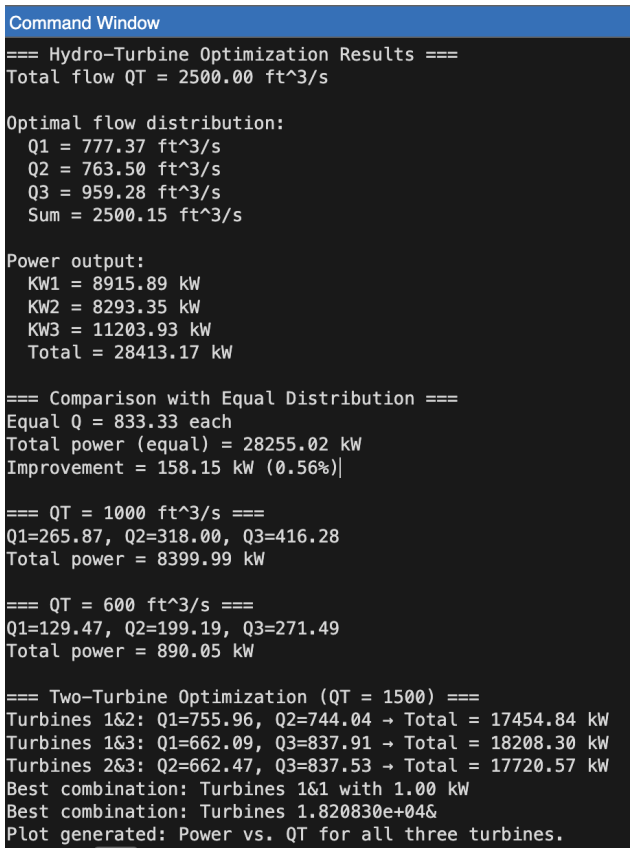
Finally, maximizing Q_3 .

$$Q_3 = .362Q_T + 54.285 \leq 1225 \quad Q_3 \leq 3231.34$$

Now we must do the inverse to get the actual bounds for each turbine. We will do minimization on Q_1 Q_2 and Q_3

Procedure 3

Now, we have a flow of 2500 feet cubed per second, we will need to distribute that between our 3 turbines, and verify that our output is the best possible maximization for what we aim to achieve. First, we'd plug in the value 2500 as QT to each turbine. Giving us $Q_1 = 777.366$, $Q_2 = 762.496$ and $Q_3 = 960.035$. Our KW1 + KW2 + KW3 = 28410.61 KW in total



```
Command Window
=== Hydro-Turbine Optimization Results ===
Total flow QT = 2500.00 ft^3/s

Optimal flow distribution:
  Q1 = 777.37 ft^3/s
  Q2 = 763.50 ft^3/s
  Q3 = 959.28 ft^3/s
  Sum = 2500.15 ft^3/s

Power output:
  KW1 = 8915.89 kW
  KW2 = 8293.35 kW
  KW3 = 11203.93 kW
  Total = 28413.17 kW

=== Comparison with Equal Distribution ===
Equal Q = 833.33 each
Total power (equal) = 28255.02 kW
Improvement = 158.15 kW (0.56%)

=== QT = 1000 ft^3/s ===
Q1=265.87, Q2=318.00, Q3=416.28
Total power = 8399.99 kW

=== QT = 600 ft^3/s ===
Q1=129.47, Q2=199.19, Q3=271.49
Total power = 890.05 kW

=== Two-Turbine Optimization (QT = 1500) ===
Turbines 1&2: Q1=755.96, Q2=744.04 → Total = 17454.84 kW
Turbines 1&3: Q1=662.09, Q3=837.91 → Total = 18208.30 kW
Turbines 2&3: Q2=662.47, Q3=837.53 → Total = 17720.57 kW
Best combination: Turbines 1&1 with 1.00 kW
Best combination: Turbines 1.820830e+04&
Plot generated: Power vs. QT for all three turbines.
```

Figure 1: MATLAB results screenshot

Procedure 4

For this problem, we'd have to simply plug in 1000 to our QT. We find that $Q_1 = 265.866$, $Q_2 = 317.505$ and $Q_3 = 416.585$. Factoring in to sum up KWH produced, we find we get 8398.207 KWH. Next we can isolate each turbine to get the power produced by an individual one. Q_1 produces 11,452.884 KWH, Q_2 produces 10843.276 KWH and Q_3 produces 12222.472 KWH.

We can also chart the power scale for 600 feet cubed per second using Desmos.

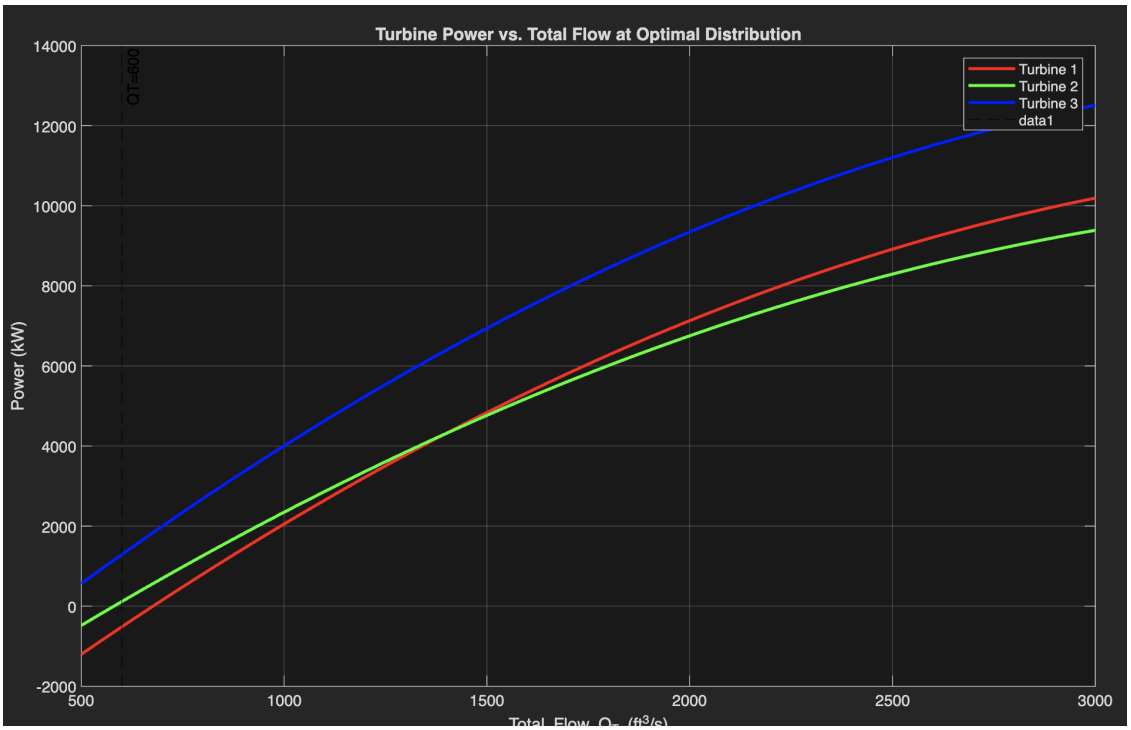


Figure 2: Power scale chart for 600 feet cubed per second

We can clearly see from this that Turbine 1, in blue, exceeds the other turbines at 600 feet cubed per second!

Procedure 5

How about using two turbines and using 1500 ft cubed, we need to find out which of the 3 would work on that flow amount. 1 and 2 interact like so $Q_2 = 86.354 + 0.870Q_1$, 2 and 3 interact like $Q_3 = 28.646 + 1.221Q_2$ and 1 and 3 interact like $Q_3 = 134.115 + 1.063Q_1$. Using this we can split the flow like so.

The first creates 17474.843, the second creates 17720.568, the last creates 18208.3. So to conclude, we would send 662.247 to Turbine 1 and 837.753 to Turbine 3 to create an optimum amount.

Procedure 6

How about if the flow is 3400 feet cubed per second? This breaks the capabilities of our turbines so we would not be able to possibly optimize the flow to work between all 3. This is impossible to know.

Conclusion

In conclusion, Lagrange multipliers are an indispensable tool with far- reaching implications in the real world. This cornerstone of calculus III proves to be much more than a mere

academic concept. Its applications span from the optimization of turbine efficiency to the intricate design of missile trajectories, demonstrating its practical utility in solving complex problems across diverse fields.

References

[1] Stewart, J., Watson, S., & Clegg, D. (2020). "Chapter 14: Partial Derivatives, Applied Project: Hydro- Turbine Optimization." In *Multivariable Calculus* (9th ed.), Cengage. Pg. 1031